# Field Extensions and Elliptic Curves

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github.com/BlockstreamResearch/zk-book

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# Plan

1 Field Extensions
- A bit of intuition
- General Definition
- Polynomial Quotient Ring
- Finite Field Extensions

2 Algebraic Closure
- Definition

3 Elliptic Curve
- Definition
- Group Structure

# Field Extensions

# $\mathbb{Q}$ vs $\mathbb{R}$

**Question #1**

What is the difference between rational numbers $\mathbb{Q}$ and real numbers $\mathbb{R}$?

**Definition**

**Rational numbers** $\mathbb{Q}$ are defined as the set $\{\frac{n}{m} : n \in \mathbb{Z}, m \in \mathbb{N}\}$.

**Question #2**

Why cannot we say $m \in \mathbb{Z}$, similarly to $n$?

**Theorem**

$\sqrt{2}$ is not a rational number. Neither is $\pi$ and $e$. But they are reals.

**Conclusion**

$\mathbb{R}$ is sort of “an extended version of $\mathbb{Q}$”.

# What about $\mathbb{R}$?

**Rhetorical Question**

Can we extend $\mathbb{R}$?

Yes — just use complex numbers $\mathbb{C}$!

**Definition**

Complex numbers $\mathbb{C}$ is defined as the set of $x + iy$ where $i^2 = -1$.

**Definition**

Complex numbers $\mathbb{C}$ are defined as the set of pairs $(x, y) \in \mathbb{R}^2$ where addition is defined as $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$, and the multiplication is:

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1).$$

# A bit about complex numbers

**Theorem**

$(\mathbb{C}, +, \times)$ is a field.

**Example**

Let us see how arithmetic is performed in $\mathbb{C}$.

- **Addition:** $(2 + 3i) + (4 + 5i) = 6 + 8i$.
- **Multiplication:** $(1 + i)(2 + i) = 2 + 3i + i^2 = 1 + 3i$.
- **Division:**

$$\frac{2+i}{1+i} = \frac{(2+i)(1-i)}{(1+i)(1-i)} = \frac{2-i-i^2}{1-i^2} = \frac{3-i}{2} = \frac{3}{2} - \frac{1}{2}i$$

**Question**

What is $(1 + i) + (2 + i)$? $i(1 + i)$? $1/i$?

# Field Extension

**Conclusion + Question**

$\mathbb{C}$ is sort of "an extended version of $\mathbb{R}$". Thus, we have

$$\mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}, \text{ where } \mathbb{Q}, \mathbb{R}, \mathbb{C} \text{ are fields}$$

So we have two questions in mind:

- Is there any mathematical term for this?
- Can we go further?

**Definition**

Let $\mathbb{F}$ be a field. A field $\mathbb{K}$ is called an **extension** of $\mathbb{F}$ if $\mathbb{F} \subset \mathbb{K}$ which we denote as $\mathbb{K}/\mathbb{F}$.

**Example**

$\mathbb{C}/\mathbb{R}$ is a field extension. So is $\mathbb{R}/\mathbb{Q}$.

# $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(i)$

## Example

Define $\mathbb{Q}(\sqrt{2}) := \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}$. This is a field extension of $\mathbb{Q}$. Arithmetic over $\mathbb{Q}(\sqrt{2})$ looks like:

- **Addition:** $(1 + 2\sqrt{2}) + (3 + 4\sqrt{2}) = 4 + 6\sqrt{2}$.
- **Multiplication:** $(1 + 2\sqrt{2})(1 + \sqrt{2}) = 1 + 3\sqrt{2} + 2\sqrt{2}^2 = 5 + 3\sqrt{2}$.
- **Division:**

$$\frac{1 + 2\sqrt{2}}{1 + \sqrt{2}} = \frac{(1 + 2\sqrt{2})(-1 + \sqrt{2})}{(1 + \sqrt{2})(-1 + \sqrt{2})} = 3 - \sqrt{2}$$

## Example

Similarly, $\mathbb{Q}(i) := \{p + qi : p, q \in \mathbb{Q}\}$ is a field extension of $\mathbb{Q}$.

# $\mathbb{Q}(\sqrt{2}, i)$

## Example

Define $\mathbb{Q}(\sqrt{2}, i) = \{\alpha + \beta\sqrt{2} : \alpha, \beta \in \mathbb{Q}(i)\}$. Typical element of $\mathbb{Q}(\sqrt{2}, i)$ can be written as:

$$(a + bi) + (c + di)\sqrt{2} = a + c\sqrt{2} + bi + d\sqrt{2}i$$

## Notice

Each element of $\mathbb{Q}(\sqrt{2}, i)$ is a linear combination of $\{1, \sqrt{2}, i, \sqrt{2}i\}$. This is usually called a **basis**. Moreover, to denote the dimensionality of $\mathbb{Q}(\sqrt{2}, i)$ over $\mathbb{Q}$, we write $[\mathbb{Q}(\sqrt{2}, i) : \mathbb{Q}] = 4$.

# Real Polynomials modulo $T^2 + 1$

**Definition. . . "Kinda"**

Consider the set $\mathcal{P}$ — a set of polynomials $\mathbb{R}[T]$ modulo $p(T) := T^2 + 1$.

**Example**

For example, $1,\ 5 + T,\ 3T,\ 1 + 2T \in \mathcal{P}$.

But what about $T^2 + 2T + 4$? We can divide by $T^2 + 1$!

$$T^2 + 2T + 4 = (T^2 + 1) \cdot 1 + (2T + 3)$$

So in $\mathcal{P}$, we have $T^2 + 2T + 4 = 2T + 3$.

# Real Polynomials modulo $T^2 + 1$

## *Arithmetic*

Over this field, we can do arithmetic as usual.

- **Addition:** $(1 + T) + (2 + 3T) = 3 + 4T$.
- **Multiplication:** $(1 + T)(2 + T) = T^2 + 3T + 2 = 3T + 1$.
- **Inverse**:

$$\left(\frac{1+T}{2}\right)^{-1} = 1 - T$$

Indeed,

$$\frac{1+T}{2} \cdot (1 - T) = \frac{1}{2} \cdot (1 - T^2) = \frac{1}{2}\left(-(T^2 + 1) + 2\right) = 1 \text{ (in } \mathcal{P})$$

# Hold on a minute...

## Results

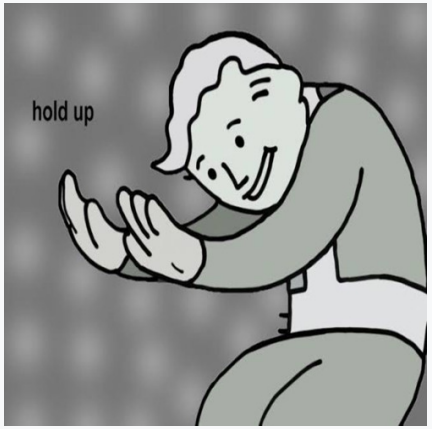
- $(1 + T) + (2 + 3T) = 3 + 4T$
- $(1 + T)(2 + T) = 1 + 3T$
- $\left(\frac{1+T}{2}\right)^{-1} = 1 - T$

## Same, but over $\mathbb{C}$

Let us do the same, but instead of $T$, use $i$.

- $(1 + i) + (2 + 3i) = 3 + 4i.$
- $(1 + i)(2 + i) = 2 + 3i + i^2 = 1 + 3i.$
- $\frac{1}{\frac{1+i}{2}} = \frac{2}{1+i} = \frac{2(1-i)}{(1+i)(1-i)} = 1 - i.$

# Hold on a minute...




So, basically, $\mathcal{P}$ and $\mathbb{C}$ have the same structure! Formally, they are isomorphic: $\mathcal{P} \cong \mathbb{C}$.

**Question**

Could we have used $T^2 + 3$ instead of $T^2 + 1$? What about $T^2 + T + 1$?

Yes, any **irreducible** 2nd-degree polynomial $p(T)$ over $\mathbb{R}$ can be used. Typically, this is denoted as $\boxed{\mathbb{R}[T]/(p)}$.

# Isomorphisms

## Reminder

For two groups $(\mathbb{G}, +)$ and $(\mathbb{H}, \times)$ we defined homomorphism to be a function $\phi : \mathbb{G} \to \mathbb{H}$ such that

$$\phi(a + b) = \phi(a) \times \phi(b)$$

However, we claim that $\mathbb{R}[T]/(T^2 + 1) \cong \mathbb{C}$, which are fields, not groups.

## Definition

A **field isomorphism** is a bijective function $\phi : (\mathbb{F}, +, \times) \to (\mathbb{K}, \oplus, \otimes)$ such that

- $\phi(a + b) = \phi(a) \oplus \phi(b)$
- $\phi(a \times b) = \phi(a) \otimes \phi(b)$
- $\phi(1_{\mathbb{F}}) = 1_{\mathbb{K}}$

But for now, $\cong$ means “exhibit the same structure”.

# Key Theorems

**Theorem**

Let $\mathbb{F}$ be a field and $\mu(T)$ — irreducible polynomial over $\mathbb{F}$ (**reduction polynomial**). Consider a set of polynomials over $\mathbb{F}[T]$ modulo $\mu$, formally denoted as $\mathbb{F}[T]/(\mu)$. Then, $\mathbb{F}[T]/(\mu)$ is a field.

**Theorem**

Let $\mathbb{F}$ be a field and $\mu \in \mathbb{F}[T]$ is an irreducible polynomial of degree $n$ and let $\mathbb{K} := \mathbb{F}[T]/(\mu)$. Let $\theta \in \mathbb{K}$ be the root of $\mu$ over $\mathbb{K}$. Then,

$$\mathbb{K} = \{c_0 + c_1\theta + \cdots + c_{n-1}\theta^{n-1} : c_0, \ldots, c_{n-1} \in \mathbb{F}\}$$

# Coming back to previous examples

**Example**

Again, consider $\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} : p, q \in \mathbb{Q}\}$. Then,

$$\mathbb{Q}(\sqrt{2}) \cong \mathbb{Q}[T]/(T^2 - 2)$$

**Example**

Similarly, $\mathbb{Q}(i) \cong \mathbb{Q}[T]/(T^2 + 1)$.

**Example**

And $\mathbb{Q}(\sqrt{2}, i)$ is just a little bit more tricky. We cannot take $(T^2 - 2)(T^2 + 1)$ since it is reducible. Instead, notice that $\theta := \sqrt{2} + i$ is a root of the irreducible polynomial

$$p(T) := T^4 - 2T^2 + 9$$

So $\mathbb{Q}(\sqrt{2}, i) \cong \mathbb{Q}[T]/(T^4 - 2T^2 + 9)$.

# Finite Field Extension

**Definition**

Recall that $\mathbb{F}_p$ (**prime field**) is a set $\{0, 1, \ldots, p-1\}$ with arithmetic modulo $p$.

In many cases, we need to extend $\mathbb{F}_p$ $2, 4, 8, 12, 24$ times. For this, we use the so-called **finite field extension**.

**Definition**

Suppose $p$ is prime and $m \geq 2$. Let $\mu \in \mathbb{F}_p[T]$ be an irreducible polynomial of degree $m$. Then, elements of $\mathbb{F}_{p^m}$ are polynomials in $\mathbb{F}_p[T]^{<m}$. In other words,

$$\mathbb{F}_{p^m} = \{c_0 + c_1 T + \cdots + c_{m-1} T^{m-1} : c_0, \ldots, c_{m-1} \in \mathbb{F}_p\},$$

where all operations are performed modulo $\mu$.

# Examples

It would be convenient to build $\mathbb{F}_{p^2}$ as $\mathbb{F}_p[i]/(i^2+1)$, but is it always possible? In other words, when $T^2 = -1$ has a solution in $\mathbb{F}_p$?

**Proposition**

Let $p$ be an odd prime. Then $T^2+1$ is irreducible in $\mathbb{F}_p[T]$ if and only if $p \equiv 3 \pmod 4$.

**Example**

Pick $p = 19$. Then $\mathbb{F}_{361} := \mathbb{F}_{19}[i]/(i^2+1)$. So typical elements are: $1+3i,\ 10+15i,\ 18+18i,\ 5,\ 7i,\ \ldots$

- **Addition:** $(1+10i)+(18+15i) = 19+25i = 6i$.
- **Multiplication:** $(5+6i)(6+7i) = 30+71i+42i^2 = -12+71i = 7+14i$.

# More Examples: Binary Extension Fields

## *Example*

Consider the $\mathbb{F}_{2^4}$. Then, there are $16$ elements in this set:

$$0, 1, T, T+1, T^2, T^2+1, T^2+T, T^2+T+1,$$
$$T^3, T^3+1, T^3+T, T^3+T+1,$$
$$T^3+T^2, T^3+T^2+1, T^3+T^2+T, T^3+T^2+T+1.$$

Set $\mu(T) := T^4 + T + 1$. Then, operations are performed in the following manner:

- **Addition:** $(T^3 + T^2 + 1) + (T^2 + T + 1) = T^3 + T$.
- **Multiplication:** $(T^3 + T^2 + 1) \cdot (T^2 + T + 1) = T^2 + 1$ since the product is $T^5 + T + 1$ and $T^5 + T + 1$ mod $(T^4 + T + 1) = T^2 + 1$.
- **Inversion:** $(T^3 + T^2 + 1)^{-1} = T^2$ since $(T^3 + T^2 + 1) \cdot T^2$ mod $(T^4 + T + 1) = 1$.

# More Examples: BN254

## *Example*

Consider the **BN254 base field**, used in SNARKs:

$$p = \text{0x30644e72e131a029} \cdots \text{a8d3c208c16d87cfd47}$$

- Then, $\mathbb{F}_{p^2} := \mathbb{F}_p[u]/(u^2 + 1)$ since $p \equiv 3 \pmod 4$.
- Define $\xi := 9 + u \in \mathbb{F}_{p^2}$. Then, set $\mathbb{F}_{p^6} := \mathbb{F}_{p^2}[v]/(v^3 - \xi)$.
- Finally, set $\mathbb{F}_{p^{12}} := \mathbb{F}_{p^6}[w]/(w^2 - v)$.

Equivalently, we can write:

$$\mathbb{F}_{p^{12}} := \mathbb{F}_p[w]/(w^{12} - 18w^6 + 82)$$

# Algebraic Closure

# Definition

**Definition**

A field $\mathbb{F}$ is called **algebraically closed** if every non-constant polynomial $p \in \mathbb{F}[T]$ has a root in $\mathbb{F}$.

**Example**

$\mathbb{R}$ is not algebraically closed since $T^2 + 1$ has no roots in $\mathbb{R}$. However, $\mathbb{C}$ is algebraically closed, which follows from the fundamental theorem of algebra. Since $\mathbb{C}$ is a field extension of $\mathbb{R}$, it is also an algebraic closure of $\mathbb{R}$. This is commonly denoted as $\overline{\mathbb{R}} = \mathbb{C}$.

**Definition**

A field $\mathbb{K}$ is called an **algebraic closure** of $\mathbb{F}$ if $\mathbb{K}/\mathbb{F}$ is algebraically closed. This is denoted as $\overline{\mathbb{F}} = \mathbb{K}$.

# Algebraic Closure for Finite Fields

Recall that we are cryptographers, not mathematicians. So we are interested in $\overline{\mathbb{F}}_p$. So I have two news to you:

- **Good news:** $\overline{\mathbb{F}}_p$ exists.
- **Bad news:** $\overline{\mathbb{F}}_p$ is infinite.

**Theorem**

No finite field $\mathbb{F}$ is algebraically closed.

**Proof.** Suppose $f_1, f_2, \ldots, f_n \in \mathbb{F}$ are all elements of $\mathbb{F}$. Consider the following polynomial:

$$p(T) := \prod_{i=1}^{n} (T - f_i) + 1 = (T - f_1)(T - f_2) \cdots (T - f_n) + 1.$$

Clearly, $p(T)$ is a non-constant polynomial and has no roots in $\mathbb{F}$, since for any $f \in \mathbb{F}$, one has $p(f) = 1$. ■

# So what?

But what form does the $\overline{\mathbb{F}}_p$ have? Well, it is a union of all $\mathbb{F}_{p^k}$ for $k \geq 1$. This is formally written as:

$$\boxed{\overline{\mathbb{F}}_p = \bigcup_{k \in \mathbb{N}} \mathbb{F}_{p^k}}$$

**Remark**

But this definition is super counter-intuitive! So here how we usually interpret it. Suppose I tell you that polynomial $q(T)$ has a root in $\overline{\mathbb{F}}_p$. What that means is that there exists some extension $\mathbb{F}_{p^m}$ such that for some $\alpha \in \mathbb{F}_{p^m}$, $q(\alpha) = 0$. We do not know how large this $m$ is, but we know that it exists. For that reason, $\overline{\mathbb{F}}_p$ is defined as an infinite union of all possible field extensions.

# Elliptic Curve

# Definition

## Definition

Suppose that $\mathbb{K}$ is a field. An **elliptic curve** $E$ over $\mathbb{K}$ is defined as a set of points $(x, y) \in \mathbb{K}^2$:

$$y^2 = x^3 + ax + b,$$

called a **Short Weierstrass equation**, where $a, b \in \mathbb{K}$ and $4a^3 + 27b^2 \neq 0$. We denote $E/\mathbb{K}$ to denote the elliptic curve over field $\mathbb{K}$.

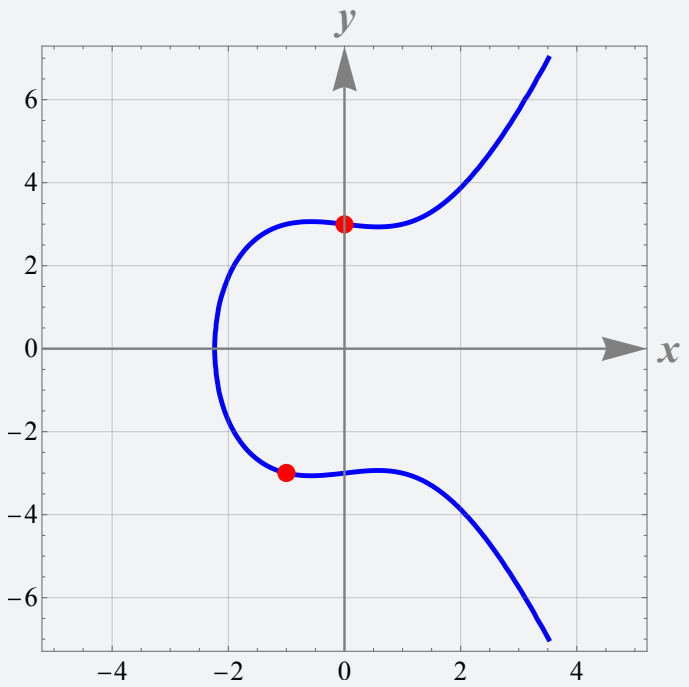
## Definition

We say that $P = (x_P, y_P) \in \mathbb{A}^2(\mathbb{K})$ is the **affine representation** of the point on the elliptic curve $E/\mathbb{K}$ if it satisfies the equation $y_P^2 = x_P^3 + ax_P + b$.

# Examples

## *Example*

Consider $E/\mathbb{Q} : y^2 = x^3 - x + 9$. Valid affine points on $E/\mathbb{Q}$ are, for example, $P = (0, 3), Q = (-1, -3) \in \mathbb{A}^2(\mathbb{Q})$.



# More Examples

Some more examples[1]:

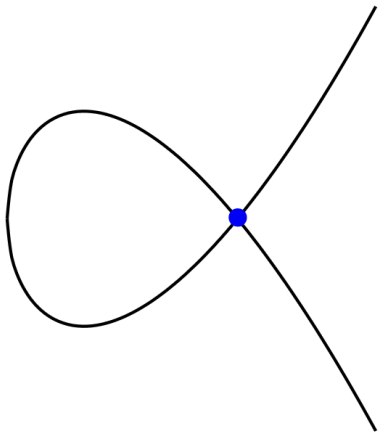


Figure 2.1: Singular curve $y^2 = x^3 - 3x + 2$ over $\mathbb{R}$.

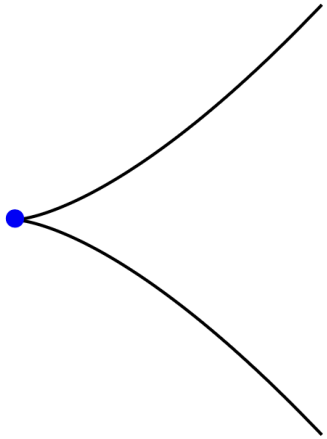


Figure 2.2: Singular curve $y^2 = x^3$ over $\mathbb{R}$.

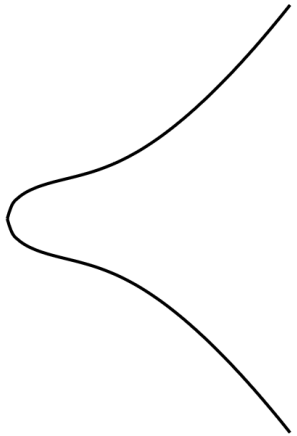


Figure 2.3: Smooth curve $y^2 = x^3 + x + 1$ over $\mathbb{R}$.

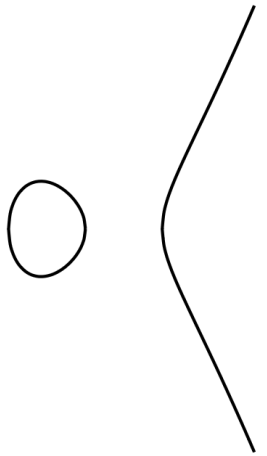


Figure 2.4: Smooth curve $y^2 = x^3 - x$ over $\mathbb{R}$.

[1] Figure taken from “Pairings for Beginners”

# Real Elliptic Curves

But real elliptic curves are not that simple. Here how they look like[2]:

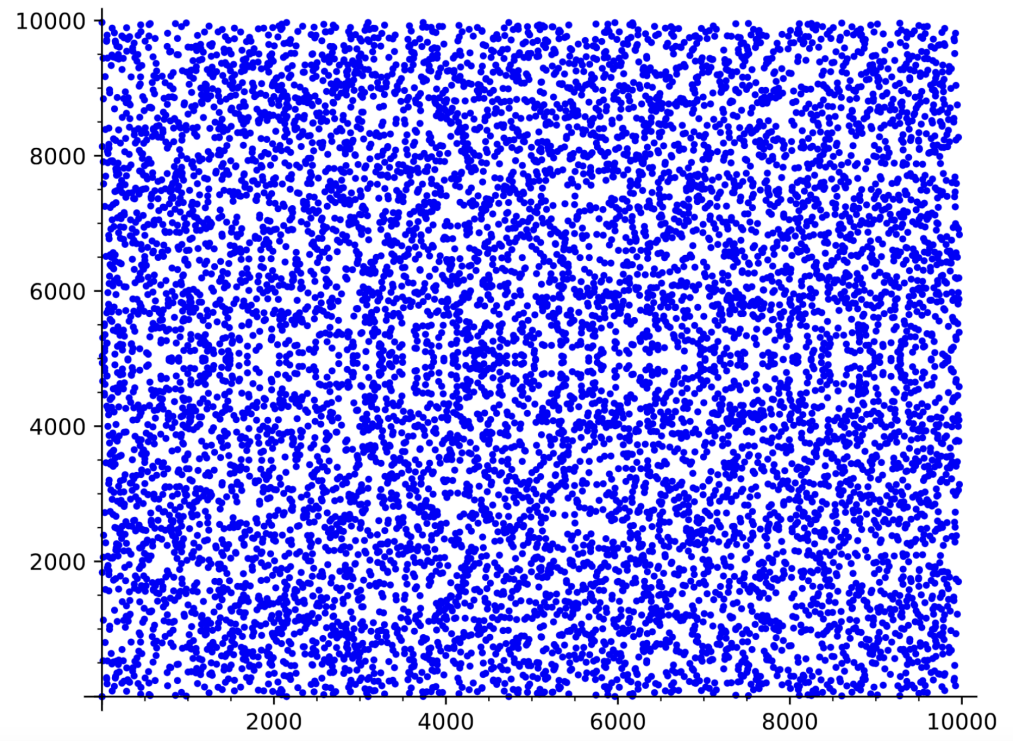



**Figure:** Curve $E/\mathbb{F}_{9973} : y^2 = x^3 - 2x + 1$ over the finite field

[2]Figure taken from “Moonmath”

# Defining a Group Structure: A Few Words

**Definition**

The set of points on the curve, denoted as $E_{a,b}(\mathbb{K})$, is defined as:

$$E_{a,b}(\mathbb{K}) = \{(x, y) \in \mathbb{A}^2(\mathbb{K}) : y^2 = x^3 + ax + b\} \cup \{\mathcal{O}\},$$

where $\mathcal{O}$ is the so-called **point at infinity**.

**Remark #1**

If $(x_P, y_P) \in E(\mathbb{K})$ then $(x_P, -y_P) \in E(\mathbb{K})$.

**Remark #2**

Typically, $\mathbb{K} = \overline{\mathbb{F}}_p$: we do not concretize over which finite field we define the elliptic curve.

# Defining a Group Structure: Chord Method

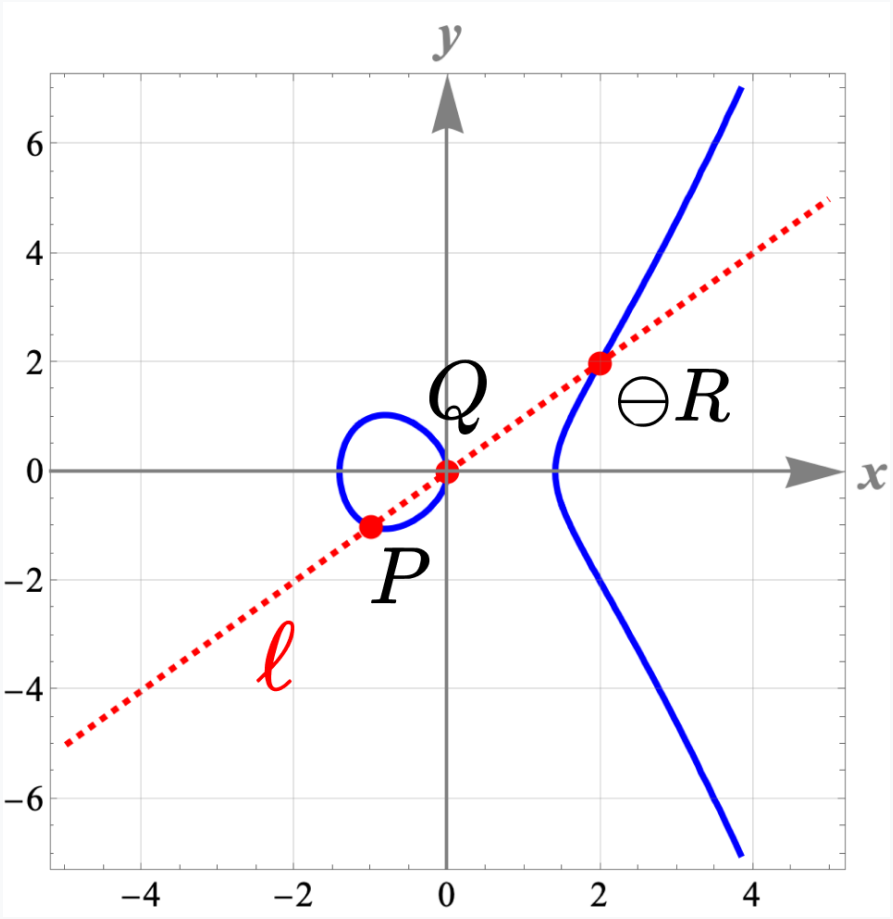



**Figure:** Chord method for adding two points

# Defining a Group Structure: Tangent Method

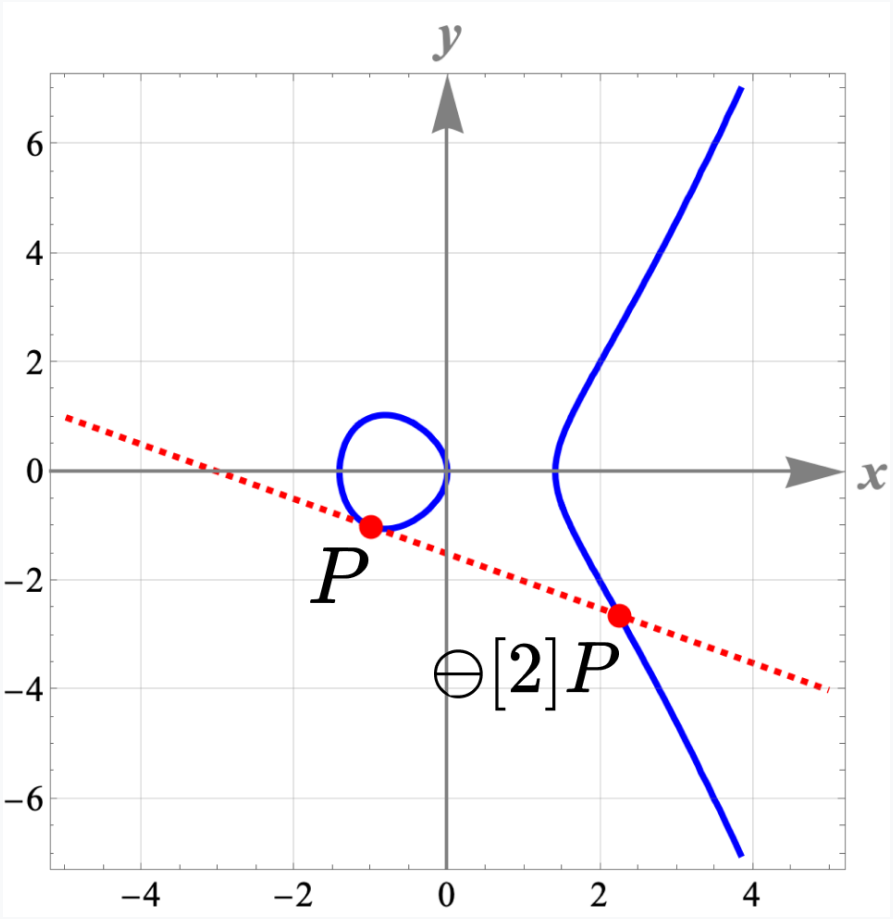



**Figure:** Tangent method for the point doubling

# Idea of Derivation

Line equation through $P = (x_P, y_P), Q = (x_Q, y_Q)$:

$$\ell : y = \lambda(x - x_P) + y_P,\ \lambda = \frac{y_Q - y_P}{x_Q - x_P}$$

So all we need is to solve the system of equations:

$$\begin{cases} y^2 = x^3 + ax + b \\ y = \lambda(x - x_P) + y_P \end{cases}$$

Substituting $y$ from the second equation to the first one, we get a cubic equation. Using Vieta's formula, one can derive

$$x_P + x_Q + x_R = \lambda^2$$

The rest is easy to finish.

# Group Law

## *Definition*

1. Point at infinity $\mathcal{O}$ is an identity element.
2. If $x_P \neq x_Q$, use the **chord method**. Define $\lambda := \frac{y_P - y_Q}{x_P - x_Q}$ — the slope between $P$ and $Q$. Set the resultant coordinates as:

$$x_R := \lambda^2 - x_P - x_Q, \quad y_R := \lambda(x_P - x_R) - y_P.$$

3. If $x_P = x_Q$ and $y_P = y_Q$ (that is, $P = Q$), use the **tangent method**. Define the slope of the tangent at $P$ as $\lambda := \frac{3x_P^2 + a}{2y_P}$ and set

$$x_R := \lambda^2 - 2x_P, \quad y_R := \lambda(x_P - x_R) - y_P.$$

4. Otherwise, define $P \oplus Q := \mathcal{O}$.

# One more Illustration

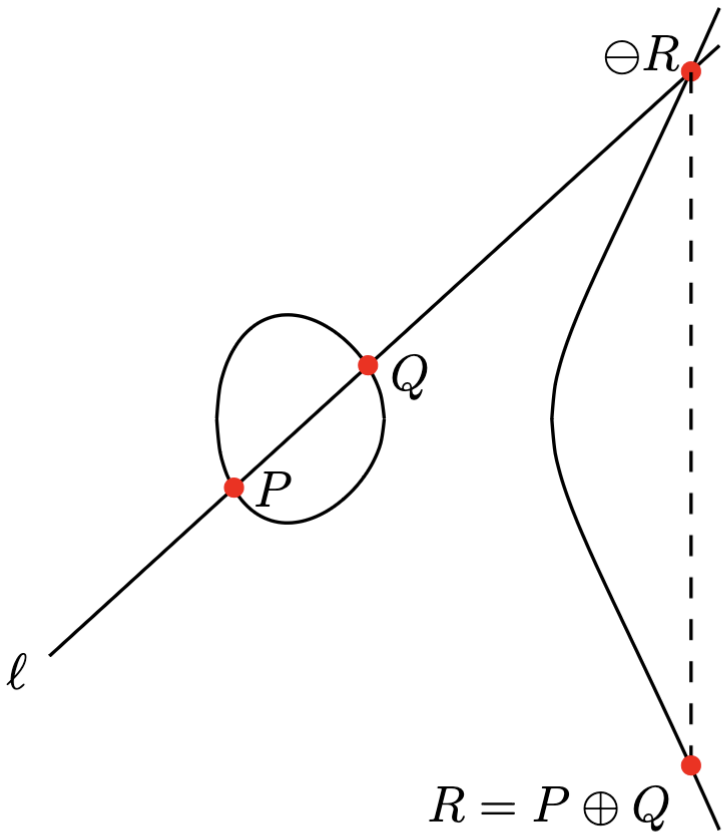



Figure 2.5: Elliptic curve addition.

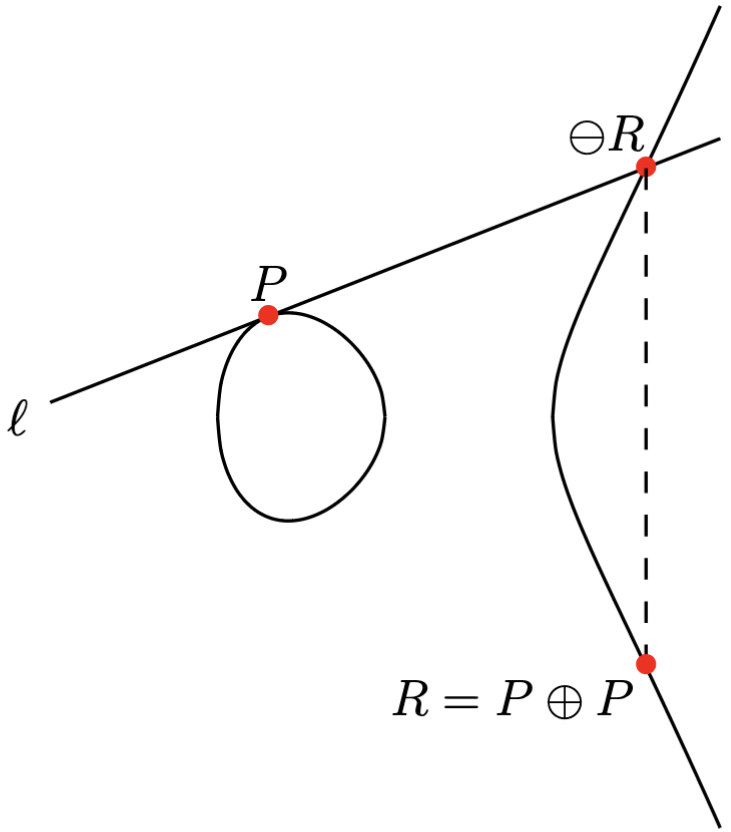


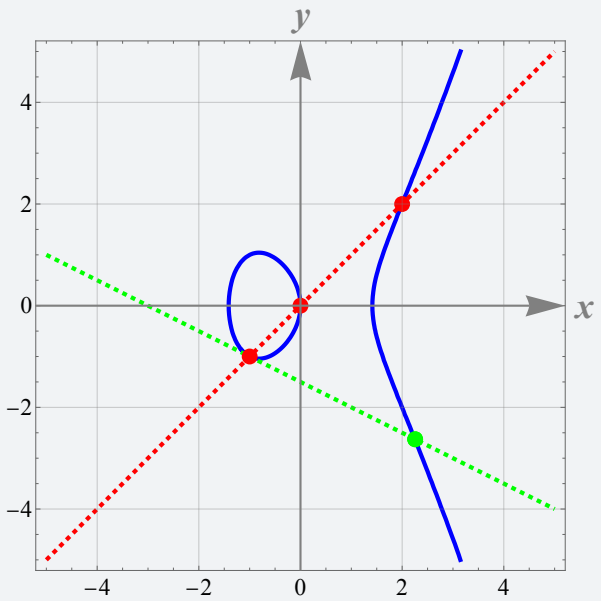

Figure 2.6: Elliptic curve doubling.

# Example

## *Example*

Consider $E/\mathbb{R} : y^2 = x^3 - 2x$.

- **Addition:** $(-1,-1) \oplus (0,0) = (2,-2),\ (2,2) \oplus (-1,-1) = (0,0).$
- **Doubling:** $[2](-1,-1) = \left(\frac{9}{4}, \frac{21}{8}\right).$



# Hasse's Theorem

**Theorem**

$(E(\mathbb{K}), \oplus)$ forms an abelian group.

Now, let us consider the group order $r := \#E(\mathbb{F}_{p^m})$.

**Theorem (Hasse's Theorem)**

$r = p^m + 1 - t$ for some integer $|t| \leq 2\sqrt{p^m}$. A bit more intuitive explanation: the number of points on the curve is close to $p^m + 1$. The value $t$ is called the **trace of Frobenius**.

**Remark**

In fact, $r = \#E(\mathbb{F}_{p^m})$ can be computed in polylogarithmic time $O(\log^\gamma(p^m))$ for some $\gamma > 1$, so the number of points can be computed efficiently even for fairly large primes $p$.

# Discrete Logarithm

**Definition**

Let $P \in E(\overline{\mathbb{F}}_p)$ and $\alpha \in \mathbb{Z}_r$. Define the scalar multiplication $[\alpha]P$ as adding $P$ to itself $\alpha - 1$ times (also set $[0]P := \mathcal{O}$).

**Definition**

Suppose $E$ is cyclic, meaning, $\langle G \rangle = E$ for some $G \in E$. The **discrete logarithm problem** on $E$ consists in the following: suppose $P = [\alpha]G$ for some $\alpha \in \mathbb{Z}_r$. Find $\alpha$ based on $P$.

**Remark**

If $r$ is a product of primes $p_1, p_2, \ldots, p_k$ such that $p_1 < p_2 < \cdots < p_k$, then the discrete logarithm problem can be solved in roughly $O(\sqrt{p_k})$: only the largest prime factor matters. For this reason, typically $r$ is prime.

# Thank you for your attention



research.blockstream.com

github.com/BlockstreamResearch/zk-book



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